Reduction of the risk of transfusion-transmitted human immunodeficiency virus (HIV) infection by using an HIV antigen/antibody combination assay in blood donation screening in Cameroon

Claude Tayou Tagny, Dora Mbanya, Lionel Leballais, Edward Murphy, Jean-Jacques Lefrère, and Syria Laperche

BACKGROUND: Improving blood safety without introducing nucleic acid testing in blood screening may be possible using antigen/antibody (Ag/Ab) combination assays, especially in resource-poor countries.

STUDY DESIGN AND METHODS: To evaluate the potential reduction of human immunodeficiency virus (HIV)-transmitted infection using such an assay, a study was carried out aimed to compare the routine strategy for blood screening (S1), combining a rapid test (Determine HIV-1/-2, Inverness) and an enzyme immunoassay (human HIV-1/-2, Human Diagnostic) with an HIV Ag/Ab assay (S2, Genscreen ULTRA HIV Ag-Ab, Bio-Rad) in 2000 blood donations tested in Cameroon. Western blot and HIV RNA polymerase chain reaction were used to confirm the infection in reactive donors, and genotype was determined in HIV RNA–positive samples.

RESULTS: Of the 2000 donors, 3.1% were positive with S1 and 4.3% with S2. Of the 97 samples positive with S1 and/or S2 tested for confirmation, 54.7% were positive, 38.1% indeterminate, and 7.2% negative. There were significant differences between S1 and S2 in terms of sensitivity (S1, 79.2%; S2, 100%), irrespective of the genotype, and in specificity (S1, 99.0; S2, 98.3%). The most frequent genotype (49%) was CRF02_AG.

CONCLUSIONS: A testing strategy using Genscreen ULTRA HIV Ag/Ab could prevent 55 HIV transmissions per 10,000 donations. However, this would be at a cost of discarding 170 per 10,000 negative donations that would test false positive, showing that the implementation of new techniques in blood screening need an optimization before routine use. Using confirmatory assays, HIV Ab prevalence in blood donors in Cameroon was estimated at 2.65%.

Despite a low prevalence rate of pathogens including human immunodeficiency virus (HIV) in their blood donor populations, many high-income countries have supplemented antibody (Ab) screening of blood donors with nucleic acid testing (NAT) to reduce the risk of transfusion-transmitted infections. Paradoxically, resource-limited countries in sub-Saharan Africa, where HIV prevalence is the highest in the world,[1-4] cannot perform NAT due to its high cost. Thus, most African blood banks perform HIV screening only with serologic tests. At least in metropolitan areas, some African blood banks use a blood screening strategy based on the use of a combination of two HIV Ab tests, according to recommendation of the World Health

ABBREVIATIONS: Ab = antibody; Ag = antigen; WB = Western blot.

From the Hematology Unit and Transfusion Service, University Teaching Hospital, Yaoundé, Cameroon; the Faculty of Medicine and Biomedical Sciences, University of Yaoundé I, Yaoundé, Cameroon; the National Reference Center for HBV, HCV, HIV in Transfusion, Institut National de Transfusion Sanguine, Paris, France; the University of California San Francisco and Blood Systems Research Institute, San Francisco, California; the Department for the Study of Blood Born Agents, Institut National de Transfusion Sanguine, Paris, France; and the Laboratory of Hematology, University Teaching Hospital, Amiens, France.

Address reprint requests to: Syria Laperche, Centre National de Référence pour les Hépatites B et C en Transfusion, Institut National de la Transfusion Sanguine, 6 rue Alexandre Cabanel, 75015 Paris, France; e-mail: slaperche@ints.fr.

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Organization (WHO). However, in remote areas, where difficult storage and transport conditions are frequently encountered, blood safety is based on the use of rapid tests that are affordable and easy to use and store and that do not need sophisticated equipment or electrical power. A recent international quality control (QC) of blood-borne virus testing, carried out in six African blood banks,[5] has demonstrated the poor sensitivity of rapid tests compared to enzyme immunoassays (EIAs) especially for the detection of hepatitis C virus (HCV) antibodies and hepatitis B antigen (Ag). For anti-HIV detection, the sensitivity of rapid tests was closer to that of EIAs. The impact of rapid test failure in blood safety is a matter of concern, particularly in areas with a high transfusion-transmitted infection prevalence rate. Improving blood safety without the use of NAT may be possible using Ag/Ab combination assays since these methods may detect window phase HCV[6-11] and HIV infection.[6,12-15] Thus, the implementation of HIV Ag/Ab combination testing may be a cost-effective strategy to reduce HIV transmission risk in developing countries.

To evaluate the benefit of introducing a HIV Ag/Ab combination assay in a high-HIV-prevalence African country, we performed a study comparing routine HIV Ab screening with a HIV Ag/Ab combination assay.

MATERIALS AND METHODS

From a total of approximately 3500 blood donations during 2008 at the Yaoundé blood center in Cameroon, we included 2000 consecutive donors in the study. Of them, 314 (15.7%) were women and 1686 (84.3%) were men with a mean age of 29.5 ± 8.7 years (range, 18-58 years). All of them were unpaid, and 79.1% were family or replacement donors.

All 2000 donors were tested in the blood bank of Yaoundé, for HIV infection with the strategy adopted for the routine blood screening (S1) and with an Ag/Ab combination assay. S1 combined a rapid HIV Type 1/HIV Type 2 (HIV-1/HIV-2) Ab assay (Determine HIV-1/-2, Inverness Medical France, Courbevoie, France) and an EIA (human enzyme-linked immunosorbent assay [ELISA] HIV 1-2, Human Diagnostic, Wiesbaden, Germany). S2 included only the Ag/Ab assay (Genscreen ULTRA HIV Ag-Ab, Bio-Rad, Marnes La Coquette, France). This assay showed 100% sensitivity for the Ab detection in a extensive independent study and has a p24 limit of detection of 13 pg/mL.[12] The two S1 assays detect HIV-1 and HIV-2 Ab only, whereas Genscreen ULTRA HIV Ag-Ab assay simultaneously detects both HIV-1 and HIV-2 Ab and p24Ag. All assays were used according to the manufacturer's instructions. For EIAs, a result was considered as positive when the signal-to-cutoff ratio was greater than or equal to 1. A sample was considered as reactive with S1 when a result was positive with one or both assays included in S1 and as reactive with S2 when assay ULTRA HIV Ag-Ab gave a positive result. To confirm that observed reactivities were due to the presence of HIV Ab (or Ag for the Ag/Ab combination assay), reactive samples with S1 and/or S2 were locally stored at –20°C until shipped in the appropriate transport conditions to the National Reference Center (Paris, France), where they were tested with Western blot (WB HIV Blot 2.2, Abbott, Rungis, France; including an HIV-2 peptide) and for HIV-1 RNA with Cobas TaqMan HIV-1 (Roche Molecular Diagnostics, Meylan, France; limit of detection, 47 copies/mL). Samples were considered as confirmed HIV positive if they were reactive on WB with at least two envelop proteins and one other protein; they were classified as negative in the absence of reactions or in the case of isolated p17 reactivity; all other patterns were considered as indeterminate. WB-positive samples were definitively considered as positive, while WB-negative and -indeterminate samples were tested for HIV-1 RNA to be classified as positive when RNA was detected and as negative when RNA was not detected (see Fig. 1).

To investigate the impact of viral diversity on the results, HIV genotype was determined in HIV RNA–positive samples, including WB-positive samples that were also tested for HIV-1 RNA with Cobas TaqMan HIV-1,

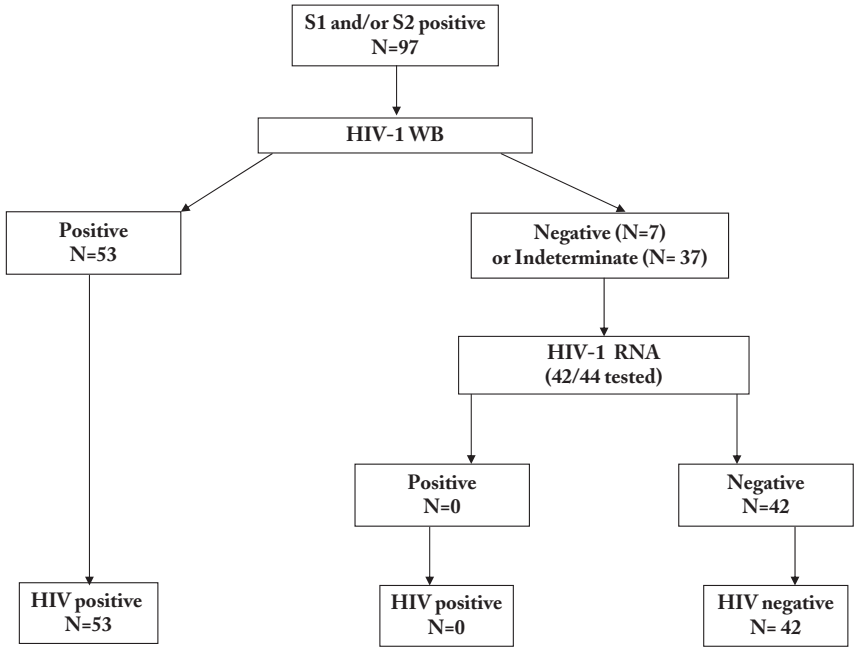


Fig. 1. Testing algorithm of reactive samples after the screening.

at the National Reference Center, by sequencing a part of the envelop gene encoding the immunodominant epitope of the transmembrane glycoprotein (gp41). The gp41 region of the envelope gene was amplified from the Cobas TaqMan RNA extraction by a nested polymerase chain reaction (PCR) protocol designed to detect phylogenetically diverse HIV variants.[16] The amplified segment was sequenced (469 nucleotides), and the genotype was determined by the neighbor-joining method. The sequence was compared with reference sequences corresponding to the nine subtypes and major circulating recombinant forms (CRF01-AE and CRF02-AG) of HIV-1 Groups M and O, retrieved from the HIV sequence database (http://hiv-web.lanl.gov). Distances were calculated using the neighbor-joining algorithm in the Phylip package based on the Kimura two-parameter distance estimation method for nucleotides Bootstrap analysis with 100 simulations was used to test the reliability of branching.

Statistical analysis was performed using the chi-square test to compare proportions or the t-test to compare means. Differences were considered significant when p values were less than 0.05.

RESULTS

Of the 2000 tested donors, 59 (2.9%) were reactive with Determine HIV-1/-2, 52 (2.6%) with human HIV-1/-2 ELISA, and 86 (4.3%) with Genscreen ULTRA HIV Ag-Ab Assay. Sixty-one (3.1%) were reactive with S1 and 97 (4.1%) with S1 and/or S2 (Table 1, Fig. 1). Of the 97 samples tested with WB, 53 (54.7%) were positive, 37 (38.1%) indeterminate, and seven (7.2%) negative according to the adopted interpretation criteria. Among the 37 indeterminate samples, 22 (59.5%) were reactive only with Genscreen ULTRA HIV Ag-Ab assay, nine (24.3%) with one or the two tests included in S1 (six with Determine, one with human HIV-1/-2 ELISA, and two with both), and six (16.2%) with S1 and S2. Thirty-five samples (94.6%) had p24 reactivity: 23 associated with one or more reactivities (14 with gp160, seven with gp160 and p17, one with p17, and one with p31), and 12 had isolated p24 reactivity. Finally, two (5.4%) samples had gp160 and p17 reactivities. Among the seven WB-negative samples, vie were reactive with Genscreen ULTRA HIV Ag-Ab assay, and two were reactive with one of the two other assays. None of the 42 tested samples (37 WB-indeterminate and five WB-negative samples available in sufficient volume), which were reactive with at least one screening assay had detectable HIV-1 RNA.

On the basis of the 53 HIV-positive donations, the sensitivity ranged from 79.2% (Determine HIV-1/-2) to 100% (Genscreen ULTRA HIV Ag-Ab) and the specificity ranged from 98.3% (Genscreen ULTRA HIV Ag-Ab) to 99.4% (human ELISA HIV-1/-2); the positive predictive value ranged from 61.6% (Genscreen ULTRA HIV Ag-Ab)

TABLE 1. Performance of HIV screening assays in the 2000 tested blood donors

Assay	Positive results (n)*				Negative results (n)†				Sensitivity¶	Specificity**	Positive predictive value (%)	Negative predictive value (%)
		WB‡				WB‡						
	Total	Pos	Ind§	Neg§	Total	Pos	Ind§	Neg§				
A: Determine HIV-1/-2	59	42	14	3	38	11	23	4	42 (79.2)	17 (99.1)	71.2	99.4
B: human ELISA HIV-1/-2	52	40	9	3	45	13	28	4	40 (75.5)	12 (99.4)	76.9	99.3
S1 = A + B	61	42	15	4	36	11	22	3	42 (79.2)	19 (99.0)	68.9	99.4
S2 = Genscreen ULTRA HIV Ag-Ab	86	53	28	5	11	0	9	2	53 (100)	33 (98.3)	61.6	100

* Ninety-seven of the 2000 donors were positive with S1 and/or S2: 47 were S1/S2 discordant and 50 were S1/S2 concordant.
† Number negative of the 97 S1 and/or S2 positive samples.
‡ Samples were considered as confirmed HIV positive (pos) if they were reactive with at least two envelop proteins and one other protein and as negative (neg) in the absence of reactions or in the case of isolated p17 reactivity; all other patterns were considered as indeterminate (ind).
§ None of the 42 tested WB indeterminate or negative tested were HIV RNA positive.
¶ Data are reported as number (%). Positive among the 53 HIV (WB and/or HIV RNA)-positive samples.
** Data are reported as number (%). False positive among the 1947 HIV-negative samples.

TABLE 2. Viral characteristics of the 53 HIV-positive samples

	S1-reactive/S2-reactive samples		S1-nonreactive/S2-reactive samples		All samples	
Genotype	Number (%)	Viral load (log copies/mL ± SD)	Number (%)	Viral load (log copies/mL ± SD)	Number (%)	Viral load (log copies/mL ± SD)
CFR02	21 (50.0)	3.93 ± 1.27	5 (45.5)	4.15 ± 0.72	26 (49)	3.97 ± 1.17
Other genotypes*	9 (21.4)	4.24 ± 0.85	2 (18.2)	3.21 ± 1.44	11 (20.8)	4.05 ± 0.94
Nongenotypable	12 (28.6)	3.23 ± 2.19	4 (36.3)	3.14 ± 1.20	16 (30.2)	3.15 ± 1.26
Total	42	3.77 ± 1.22	11	3.74 ± 1.13	53	3.77 ± 1.19

* Of 11 non-CRF02, four were D, two CRF06, one CRF11, one CRF13, one CRF22 (negative with S1), oneF2, and one G.

to 76.9% (human ELISA HIV-1/-2) and the negative predictive value ranged from 99.3% (human ELISA HIV-1/-2) to 100% (Genscreen ULTRA HIV Ag-Ab). There was a significant difference between S1 and S2 for both sensitivity ($p < 10^{-3}$; S1, 79.2%; and S2, 100%) and specificity ($p = 0.05$; S1, 99.0; and S2, 98.3%). WB patterns seen for the 11 S2 only–reactive samples were more consistent with chronic than early-phase HIV infection. In addition, there was no difference in the mean viral loads found in the 11 discordant samples (3.74 ± 1.13 log IU/mL; range, 1.67-5.94 log IU/mL) compared to the 42 concordant samples (3.77 ± 1.22 log IU/mL; range, 0-5.18 log IU/mL), irrespective of the genotype (Table 2).

The HIV genotypes of the 53 positive samples are summarized in Table 2: 16 (30.2%) were not typable, 26 (49%) were CRF02_AG, and the remaining 11 (20.8%) were distributed as follows: four were D, two CRF06_cpx, one CRF11_cpx, one CRF13_cpx, one CRF22_cpx, one F2, and one G. There was no difference in this distribution between the 42 S1/S2 concordant samples (CRF02, 62%; other genotypes, 19%; 19% not typable) and the 11 S1/S2 discordant samples (CRF02, 46%; other genotypes, 18%; 36% not typable).

DISCUSSION

Because of the high HIV prevalence in the general population of Cameroon, currently estimated at 5.5%,[17,18] the University Teaching Hospital of Yaoundé adapted the HIV screening strategy recommended by the WHO, namely, one rapid HIV Ab test and one HIV Ab ELISA. The results obtained in this study indicate that EIA testing (human ELISA HIV-1/-2) did not provide any added value when compared to the rapid assay (Determine HIV-1/-2) when used in a strategy by which a positive result on either assay was sufficient to discard the donation. Indeed, the negative predictive value of S1 was the same as the Determine assay alone, and the Human ELISA HIV-1/-2 had a lower sensitivity than Determine. The lower sensitivity of rapid tests,[19] due to the low Ab titer[20-22] or to the viral diversity,[23,24] has been reported as a limitation when compared to EIAs. This was not the case in our study, which suggests that HIV rapid tests could reach at level of performance equivalent to some EIAs. Thus blood screening with rapid tests may remain a viable screening option in hospitals with basic laboratories and a limited blood supply, particularly when emergency transfusion situations need rapid screening.

Our findings clearly demonstrate that the systematic use of two assays for the screening of HIV infection in blood donors is questionable when the second assay is a screening rather than a confirmatory assay and particularly in the case when it has a low sensitivity. If a blood screening strategy is based on the use of only one assay it must obviously be the best assay available. Many studies[12,14,15,25-28] have reported the benefit of the use of combination Ag/Ab assays in high HIV prevalence and incidence areas, since they are able to detect infected blood donors in the viremic preseroconversion window period. In our study, Genscreen ULTRA HIV Ag-Ab assay exhibited the highest sensitivity when compared to the routinely used assays. However, we were surprised that the 11 false-negative results with S1 did not appear to be early HIV infections according to the WB and PCR data, nor could these results be attributed to viral diversity, since the distribution of HIV subtypes was similar in the limited number of false-negative and true-positive S1 samples. Despite satisfactory results obtained by the Yaoundé Blood Bank in a recent QC study,[5] we cannot exclude errors in technique in relationship, for example, with inaccurate machine adjustments as an explanation for the current findings. Although this hypothesis could explain negative results obtained with one assay, especially the subjective rapid test, this would be unlikely to explain negative results on both assays. Unfortunately, there is no residual sample volume for further investigations.

The Genscreen ULTRA HIV Ag-Ab assay had a very poor specificity (98.3%) in this study, less than that required for the CE mark (99.5%). The false-positive rate may have been inflated since all initially positive samples counted in the calculation of specificity and were not retested. Furthermore, Genscreen ULTRA HIV Ag-Ab assay, recently introduced in the Yaoundé blood bank probably needs some technical adjustments to reduce the number of false-positive results. On the other hand, we may have underestimated HIV infection in donors with indeterminate or negative results due to the stringent WB

criteria used in this study, which may have missed early HIV infections. This hypothesis is less likely because all negative and indeterminate samples were HIV-1 RNA negative with Cobas Taq Man HIV-1, as well as the eight randomly selected nonpositive samples (six WB indeterminate from which two were positive with all assays and two WB negative) also tested with the nested PCR used for the genotyping. We did this because false-negative results have been described with commercial NAT assays,[29] particularly for some CRF_02 strains, which may be missed by Cobas TaqMan HIV-1.[30] Even though we cannot totally eliminate the presence of some variants including CRF_02 not detected by the NAT methods used in our study, viral diversity is unlikely to explain the high number of indeterminate samples (38.1%). Interestingly, indeterminate profiles were mainly obtained from p24 protein, as observed in a study carried out in Ethiopian population.[31] Such WB patterns may be explained by cross-reactivity to other infections seen frequently in Africa including tuberculosis,[32] leprosy,[33] and malaria.[34-36] Although it would be negligible in clinical practice, the level of specificity we observed is unacceptable for blood screening because it leads to discarding many blood units. This is particularly important in resource-limited countries, where the number of units of blood collected per 1000 inhabitants ranges from 1 to 15 units compared to 30 to 75 in developed countries).[37]

The testing algorithm adopted in this study, particularly the inclusion of confirmatory assays, led to a more accurate estimate of HIV Ab prevalence in Cameroon blood donors. The prevalence of 2.65% is similar to that previously reported in the same blood center in 2004, based on two rapid tests (2.9%).[38] However, including only samples confirmed positive with confirmatory tests is the best indicator of the prevalence of HIV-infected donors since it excludes falsely positive and falsely negative samples.

The viral diversity observed in our study is in accordance with those recently reported in Cameroon[39] with a dominance of CRF02_AG, representing 49% of the positive blood donors investigated for genotype. Notably, neither HIV-1 Group O nor HIV-2, generally observed in Cameroon,[40] was identified. However, the number of donors with HIV genotypes was too low to allow precise estimates of the distribution of HIV genotypes circulating in HIV Cameroonian blood donors. In addition, genotyping on a single, short env sequence does not permit firm conclusions on the precise genotype, especially owing to high diversity and frequent recombination.

We have demonstrated that the high sensitivity of Genscreen ULTRA HIV Ag-Ab assay would allow the prevention of 55 HIV transmissions in 10,000 tested donations. Extrapolating these results to the 70,000 donations collected per year in Cameroon, 385 infections could be avoided each year if Ag/Ab combined testing could be carried out across the whole country. However, this would be at a cost of discarding 170 per 10,000 negative donations that would test false positive due to the poor specificity of this assay. These findings highlight that the decision to implement new blood screening strategies must take local conditions into account. We do not believe that viral diversity affected the current results and no substantial changes in the diversity of HIV-1 strains circulating in the Cameroonian blood donor population have been noted.[39] Nevertheless, ongoing surveillance for newly emerging HIV variants is essential if screening tools are to be adapted for better blood safety.

The results of this study should encourage health authorities in sub-Saharan countries to consider new strategies for HIV screening of blood donations using highly sensitive tests such as Ag/Ab combination assays to reduce the risk of HIV infection from transfusion. However, the National Blood Center in each developing country should first estimate the benefit of implementing such a strategy in its own blood screening policy.

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CONFLICT OF INTEREST

The authors declare no conflict of interest associated with this study.

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